Carl E. Hjelmen, PhD

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PROFESSIONAL POSITIONS

2026–Pres. Associate Professor, Dept. of Biology, Utah Valley University, Orem, UT
Bioinformatics Degree Program Coordinator
Hiring Committee Chair

2021–2026 Assistant Professor, Dept. of Biology, Utah Valley University, Orem, UT

EDUCATION, RESEARCH, AND TRAINING

2019–2021 Postdoctoral Research Associate, Dept. of Biology, Texas A&M University, College Station, TX
PI: Heath Blackmon

2017–2018 Postdoctoral Research Associate, Dept. of Entomology, Texas A&M University, College Station, TX
PI: Aaron Tarone
Project: Proteomic and microRNA markers of fly development

Aug. 2017 Ph.D., Entomology, Texas A&M University, College Station, TX
Advisor: J. Spencer Johnston
Dissertation: Phylogenetic analyses of genome size evolution in Drosophilidae

May 2013 B.A. in Biology, Augustana College, Sioux Falls, SD

SCHOLARSHIP

Reviewed Publications (☨ = Mentored Student; ! = Undergraduate Author)

28. Sylvester T, Hjelmen CE, Thompson MJ, Blackmon LT, Alfieri JM, Hoover Z, Esfandani T, Blackmon H. Population Structure of the Jewel Scarab *Chrysina gloriosa*. *Genome Biology and Evolution*. doi: 10.1093/gbe/evag123
27. Abair A, Egan AN, ☨!Bugg B, et al., Hjelmen CE, Bailey CD. Phylotranscriptomics and genome size evolution in *Leucaena* (Fabaceae). *American Journal of Botany*. doi: 10.1002/ajb2.70178
26. Zahn G, Hjelmen CE, Wainwright B. Distinct temporal trajectories of bacterial and fungal networks during agricultural rewilding. *Science of the Total Environment*. doi: 10.1016/j.scitotenv.2025.180999
25. ☨!Motte RR, Hjelmen CE. DSCAM1 evolution and its relationship to sociality: Hymenopteran DSCAM1 exhibits accelerated evolution in variable exon regions. *Insect Systematics and Diversity*. doi.org/10.1093/isd/ixaf031
24. Oakes J, Kuddus JN, Downs E, Oakey C, Davis K, Mohammad L, Whitely K, Hjelmen CE, Kuddus R. Isolation and Characterization of a Crude Oil-Tolerant Obligate Halophilic Bacterium from the Great Salt Lake. *Microorganisms*. doi.org/10.3390/microorganisms13071568
23. Copeland M, et al., Hjelmen CE, Rieski LK, Blackmon H, Casola C. (2024). Genome assembly of the southern pine beetle reveals the origins of gene content reduction in *Dendroctonus*. *R. Soc. Open Sci*. 11: 240755. doi:10.1098/rsos.240755
22. Hjelmen CE. (2024). Genome size and chromosome number are critical metrics for accurate genome assembly assessment in Eukaryota. *Genetics*. doi: 10.1093/genetics/iyae099
21. Sylvester T, Hoover Z, Hjelmen CE, et al., Blackmon H. (2024). A reference quality genome assembly for the Jewel scarab Chrysina gloriosa. *Genes, Genomes, Genetics*. doi: 10.1093/g3journal/jkae084
20. Hjelmen CE, Yuan Y, Parrot JJ, !McGuane AS, Srivastav SP, Purcell AC, Pimsler ML, Sze S-H, Tarone AM. (2022). Identification and Characterization of Small RNA Markers of Age in the Blow Fly *Cochliomyia macellaria*. *Insects*. doi.org/10.3390/insects13100948

19. Morelli MW†!, Blackmon H, Hjelmen CE. (2022). Diptera and *Drosophila* Karyotype Databases. *Frontiers in Ecology and Evolution*. doi: 10.3389/fevo.2022.832378
18. Pimsler ML, Hjelmen CE, Jonika MM, et al. (2021). Sexual dimorphism in growth rate and gene expression *in Chrysomya rufifacies*. *Frontiers in Ecology and Evolution*, 9:696638. doi: 10.3389/fevo.2021.696638
17. Tvedte ES, et al., Hjelmen CE, Johnston JS, et al. (2021). Comparison of long-read sequencing technologies in interrogating bacteria and fly genomes. *G3: Genes, Genomes, Genetics*. doi: 10.1093/g3journal/jkab083
16. !Anderson NW, Hjelmen CE, Blackmon H. (2020). The probability of fusions joining sex chromosomes and autosomes*. Biology Letters*. doi: 10.1098/rsbl.2020.0648
15. Sylvester TP, Hjelmen CE, Hanrahan SJ, Lehnart PA, Johnston JS, Blackmon H. (2020). Lineage-specific patterns of chromosome evolution in Polyneopteran insects. *Proc. R. Soc. B*, 287: 20201388. doi: 10.1098/rspb.2020.1388
14. Malawey AS, et al., Hjelmen CE, et al. (2020). Interaction of age and temperature on heat shock protein expression in the Black Soldier Fly. *Journal of Insects as Food and Feed*. doi: 10.3920/JIFF2020.0017
13. Hjelmen CE, †Holmes VR, !Burrus CG, †!Piron E, †!Mynes M, †!Garrett M, Blackmon H, Johnston JS. (2020). Thoracic underreplication in *Drosophila* estimates a minimum genome size. *Evolution*. doi: 10.1111/evo.14022
12. †!Jonika MM, Hjelmen CE, Faris AM, †!McGuane AS, Tarone AM. (2020). Differentially spliced genes as markers for sex for forensic entomology. *Journal of Forensic Sciences*. doi: 10.1111/1556-4029.14461
11. Johnston JS, !Zapalac ME, Hjelmen CE. (2020). Flying High—Muscle specific underreplication in *Drosophila*. *Genes* 11, 246. doi: 10.3390/genes11030246
10. Hjelmen CE, Parrott JJ, Srivastav S, †!McGuane AS, et al. (2020). Effect of phenotype selection on genome size variation in two species of Diptera. *Genes* 11, 218. doi: 10.3390/genes11020218
9. Hjelmen CE, Blackmon H, †Holmes VR, !Burrus CG, Johnston JS. (2019). Genome size evolution differs between *Drosophila* subgenera. *G3: Genes, Genomes, Genetics* 9(10): 3167–3179
8. Hjelmen CE, †!Garrett M, †!Holmes VR, †!Mynes M, †!Piron E, Johnston JS. (2018). Genome size evolution within and between the sexes. *Journal of Heredity*. 110(2): 219–228
7. Lower SS, Johnston JS, Stanger-Hall K, Hjelmen CE, et al. (2017). Genome size in North American fireflies: Substantial variation likely driven by neutral processes. *Genome Biol. Evol*. evx097
6. Hjelmen CE and Johnston JS. (2017). The mode and tempo of genome size evolution in the subgenus *Sophophora*. *PLOS One* 12(3), e0173505
5. Arnqvist G, Sayadi A, Immonen E, Hotzy C, Rankin D, Tuda M, Hjelmen CE, Johnston JS. (2015). Genome size correlates with reproductive fitness in seed beetles*. Proc. R. Soc. B*. 282(1815)
4. Rangel J, !Strauss K, !Seedorf K, Hjelmen CE, Johnston JS. (2015). Endopolyploidy changes with age-related polyethism in the honey bee. *PLoS One*, 10(4), e0122208
3. Ellis LL, et al., Hjelmen CE, Moore KL, Mackay TFC, Johnston JS, Tarone AM. (2014). Intrapopulation Genome Size Variation in *D. melanogaster*. *PLoS Genet* 10(7): e1004522
2. The DGRP Consortium. (2014). Natural variation in genome architecture among 205 *D. melanogaster* Genetic Reference Panel lines. *Genome Research*: 1193–1208
1. Larson MK, Tormoen GW, Weaver LJ, Luepke KJ, Patel IA, Hjelmen CE, Ensz NM, McComas LS, McCarty OJ. (2012). Exogenous modification of platelet membranes with EPA and DHA reduces platelet procoagulant activity. *Am J Physiol Cell Physiol* 304(3):273

In Review and Preparation

- †!Curnow SP, Olson M, Hjelmen CE. Unraveling the Enigma: *Drosophila* Genome Size Evolution and Its Potential Environmental Correlates. [Submitted]
- Hjelmen CE, Fout S†!, Parrott JJ, McGuane AS, Jonika MM, Purcell AC, Srivastav S, Sing-Hoi S, Tarone AM. Characterization of miRNA in four forensically relevant species of Diptera. [In prep / data analysis]

Book Chapters

- Johnston JS, Bernardini A, Hjelmen CE. (2019). Genome Size Estimation and Quantitative Cytogenetics in Insects. In: Insect Genomics: Methods and Protocols.

Proceedings Papers

- † Jonika MM, Faris AM, Hjelmen CE, Tarone AM. (2019). Transcriptional Markers of Sex Determination for Forensic Entomology. Proceedings of the American Academy of Forensic Sciences, 71st Annual Scientific Meeting.

Other

- Comparative phylogenetic lab activity on phylogenetic independent contrasts and PGLS for Spring 2016 ENTO 606 (Quantitative Phylogenetics).

SEMINARS AND CONFERENCE TALKS

(*Presenting Author; † = Mentored Student; ! = Undergraduate Author)

2026

A New Look at Old Questions: Insights from an Undergraduate-Centered Lab. *Hjelmen CE.
Genomics Education Partnership, Online

A New Look at Old Questions. *Hjelmen CE.
Georgia State University–Perimeter EXCITE Spring Symposium Keynote, Online

2025

It's entomology, not etymology: Activities to solidify students' understanding of entomological terms. *Hjelmen CE.
Entomological Society of America Annual Meeting, Portland, OR

Circling Back with Fresh Eyes: Rethinking Genome Evolution in a Student-Powered Lab. *Hjelmen CE.
Dept. of Entomology and Plant Pathology Seminar Series, Oklahoma State University, Stillwater, OK

2024

Evolution of genome size and structures in the *Drosophila* genus. *Hjelmen CE.
Utah *Drosophila* Fall Symposium, University of Utah, Salt Lake City, UT

Bugs in the system: A computational biologist's first semester teaching entomology. *Hjelmen CE.
Entomological Society of America Annual Meeting, Phoenix, AZ

Insect Genome Size Evolution: What Do We Know and Where Do We Go from Here? *Hjelmen CE.
International Plant and Animal Genome Conference, San Diego, CA

2023

Implementing bioinformatics and biotechnology in the classroom. *Hjelmen CE.
Entomological Society of America Annual Meeting, National Harbor, MD

A Tale of the other guy: The story of Alfred Russel Wallace. *Hjelmen CE.
UVU, Dept. of Biology Colloquium Series

Developing open-source databases for facilitating undergraduate research. *Hjelmen CE.
Bioinformatics and Computational Biology in the Liberal Arts Workshop, Reed College, Portland, OR

Application of Phylogenetic Analyses. *Hjelmen CE. [Virtual]
Annual North American Forensic Entomology Association Meeting, Arizona State University

Sifting through the mess of heterochromatin: Assembling heterochromatic *Drosophila* genomes. !Ohran M, *Hjelmen CE.
Pacific Branch of ESA Annual Meeting, Seattle, WA

2022

Student driven data mining for freely available databases and 'free' research projects. *Hjelmen CE.
Entomological Society of America Annual Meeting, Vancouver, Canada

Come Together: Chemistry of the beetles—A story of acid and flashing lights. *Hjelmen CE.
Central Utah Section of the American Chemical Society, Earth Day Seminar

So, you want to go to graduate school… *Hjelmen CE.
UVU, Dept. of Biology Colloquium Series

2021

Working out the 'bugs' in genome size evolution. *Hjelmen CE.
UVU, Dept. of Biology Colloquium Series

2020

Working out the 'bugs' in genome size evolution. *Hjelmen CE.
Rutgers University, Dept. of Entomology Invited Seminar

2019

What does the rate of change in heterochromatin tell us about genome size evolution and the minimum DNA content of a fly? *Hjelmen CE, Johnston JS, Blackmon H.
Biology Dept. Student Postdoc Research Conference, Texas A&M University

How much DNA does it take to be a fly and what happens to the rest? *Hjelmen CE, Johnston JS, Blackmon H.
Ecology and Evolutionary Biology Seminar Series, Texas A&M University, College Station, TX

2018

Differential expression of proteins in species of forensically relevant Diptera. *Hjelmen CE, Srivastav S, Parrott JJ, Dangott LJ, Tarone AM.
Entomological Society of America Annual Meeting, Vancouver, Canada

Differential expression of proteins. *Hjelmen CE, et al.
Annual NAFEA Meeting, Orlando, FL

What amount of DNA is just right? *Hjelmen CE, ☨Holmes VR, Johnston JS.
Southeast Texas Evolutionary Genetics and Genomics Meeting, Houston, TX

2017

What is underreplication and how does this phenomenon contribute to the enigma of genome size evolution in *Drosophila*? *Hjelmen CE and Johnston JS.
ESA Southwestern Branch Meeting, Austin, TX (1st place)

2016

What's the buzz in *Drosophila* genome size: A phylogenetic comparison of *Sophophora* and *Drosophila*. *Hjelmen CE and Johnston JS.
International Congress of Entomology, Orlando, FL

How does replication level contribute to genome size evolution in *Drosophila*? *Hjelmen CE and Johnston JS.
19th Annual Graduate Student Forum, Dept. of Entomology, Texas A&M University

How do temperature differences relate to genome size variation? *Hjelmen CE and Johnston JS.
Ecological Integration Symposium / Student Research Week, Texas A&M University; ESA Southwestern Branch Meeting, Tyler, TX

2015

Phylogenetic basis for understanding genome size evolution in *Drosophila*. *Hjelmen CE and Johnston JS.
ESA Annual Meeting, Minneapolis, MN (1st place); 18th Annual Graduate Student Forum, Texas A&M; Ecological Integration Symposium (1st place); Student Research Week (1st place)

2012

Beyond the Augie Borders: Student Research Off Campus (Texas A&M University). *Hjelmen CE.
Augustana College Biology Dept.—Invited Oral Presentation

Collaborator and Student Talks

2025

What flies eat and excrete: Insights from puke, poop, and gut contents of blow flies. *Weidner LM, Morales C, ☨!Jannuzzi B, et al., Hjelmen CE.
ESA Annual Meeting, Portland, OR

Enhancing entomology education by creating online resources and upgrading a teaching collection. !*☨Nielsen C, Rotter M, Hjelmen CE.

ESA Annual Meeting, Portland, OR (1st Place)

Optimizing bioinformatic pipelines for profiling blow fly diets. !*†Jannuzzi B, Meeds A, Weidner LM, Hjelmen CE.
ESA Annual Meeting, Portland, OR (2nd Place)

Genome size evolution in Hymenoptera: Phylogenetic insights into social structure. !*†Frary O, !†Jetton B, Hjelmen CE.
Pacific Branch of ESA Annual Meeting, Salt Lake City, UT

Blow fly biodiversity and ecological roles in Utah's diverse landscapes. !*Beck H, Weidner LM, Hjelmen CE.
Pacific Branch of ESA Annual Meeting, Salt Lake City, UT

Unpacking chromosomal evolution in Diptera. !*Jetton B, Hjelmen CE.
Pacific Branch of ESA Annual Meeting, Salt Lake City, UT

2024

A phylogenetic analysis of genome size evolution and social structure in Hymenoptera. !*Frary O, Hjelmen CE.
ESA Annual Meeting, Phoenix, AZ

Counting chromosomes: Exploring evolutionary insights from *Drosophila* karyotypes. !*French A, Hjelmen CE.
ESA Annual Meeting, Phoenix, AZ

The 'fly'-logeny of *Drosophila* Chromosome Evolution. !*French A, Hjelmen CE.
Utah Conference on Undergraduate Research 2024, Utah Valley University

Accelerated Rates of Evolution in hymenopteran DSCAM genes. !*Motte R, Hjelmen CE.
Utah Conference on Undergraduate Research 2024, Utah Valley University

2023

The evolution of an undergraduate independent research project. !*Motte R, Hjelmen CE.
ESA Annual Meeting, National Harbor, MD

Sifting through the mess of heterochromatin: Assembling heterochromatic *Drosophila* genomes. !*Ohran M, Hjelmen CE.
UVU Showcase, Orem, UT

Investigating the relationship between DSCAM evolution and arthropod sociality. !*Motte RR, Hjelmen CE.
Pacific Branch of ESA Annual Meeting, Seattle, WA

2020

An evaluation of differentially spliced genes as markers of sex for forensic entomology. *†Jonika MM, Hjelmen CE, Faris AM, McGuane AS, Tarone AM.
Annual Meeting of the North American Forensic Entomology Association

2019

Transcriptional Markers of Sex Determination for Forensic Entomology. *†Jonika MM, Faris AM, Hjelmen CE, Tarone AM.
American Academy of Forensic Sciences Annual Meeting, Baltimore, MD

2018

Transcript-based sex determination for forensic entomology. *†!Jonika MM, Faris AM, Hjelmen CE, Tarone AM.
ESA Annual Meeting, Vancouver, Canada; Ecological Integration Symposium, Texas A&M; Student Research Week, Texas A&M

Poster Presentations

2025

From city to canyon: A survey of blow fly biodiversity across Utah ecoregions. *†!Beck H, †!Jannuzzi B, Weidner LM, Hjelmen CE.
ESA Annual Meeting, Portland, OR (1st Place)

Oxford Nanopore Technology sequencing and identification of fly gut content. *†!Jannuzzi B, Meeds A, Weidner LM, Hjelmen CE.
Pacific Branch of ESA Annual Meeting, Salt Lake City, UT

2024

Decoding Diptera: Unraveling chromosome evolution across fly families. *†!Jetton B, Hjelmen CE.
ESA Annual Meeting, Phoenix, AZ

Preliminary survey of blow fly species (Diptera: Calliphoridae) across Utah. *†!Beck H, Hjelmen CE.
ESA Annual Meeting, Phoenix, AZ

Do the differences in size between heteromorphic sex chromosomes influence organism longevity? *†!Frary O, Hjelmen CE.
Utah Conference on Undergraduate Research 2024

Creating a universal framework for reconstructing phylogenies using R. *†!Jetton B, Hjelmen CE.
Utah Conference on Undergraduate Research 2024

Survey of Blow Fly Species Across Utah and Salt Lake Counties. *†!Beck H, Weidner LM, Hjelmen CE.
Utah Conference on Undergraduate Research 2024

Investigating the relationship between natural environment and drosophilid genome size. *†!Curnow S, Hjelmen CE.
Utah Conference on Undergraduate Research 2024

2023

Morphometrics of Jewel Scarabs of the Southwest. *Blackmon LT, Jonika MM, Sylvester T, Alfieri J, Hjelmen CE, Blackmon H.
Evolution Annual Meeting, Albuquerque, NM

Time flies: Chromosomes number changes in the evolutionary history of *Drosophila*. *†!French A, Hjelmen CE.
UVU Showcase; Utah Conference on Undergraduate Research 2023, University of Utah

Comparing highly heterochromatic cactophilic *Drosophila* genome assemblies. *†!Ohran M, Hjelmen CE.
Utah Conference on Undergraduate Research 2023

Environmental factors have little influence on drosophilid genome size. *†!Curnow S, Hjelmen CE.
Utah Conference on Undergraduate Research 2023

Smart animals and social critters: DSCAM evolution and sociality. *†!Motte RR, Hjelmen CE.
Utah Conference on Undergraduate Research 2023

2022

Assembly of highly heterochromatic cactophilic *Drosophila* genomes. *†!Ohran M, Hjelmen CE.
ESA Annual Meeting, Vancouver, Canada (2nd Place)

Do environmental factors impact drosophilid genome size evolution? *†!Curnow S, Hjelmen CE.
ESA Annual Meeting, Vancouver, Canada

2020

Heterochromatin profiles and sex chromosomes of desert *Drosophila*. *Hjelmen CE.
Arthropod Genomics 2020—Virtual meeting

Rates of Change in Heterochromatin and Estimating the Minimum Genome Size of a Fly. *Hjelmen CE, †Holmes VR, !Burrus CG, †!Piron E, †!Mynes M, †!Garrett MA, Blackmon H, Johnston JS.
Life Sciences PhD Recruiting Event, Texas A&M University

2019

Genome size evolution differs between *Drosophila* subgenera. *Hjelmen CE, Blackmon H, Holmes VR, !Burrus CG, Johnston JS.
Society for the Study of Evolution Annual Meeting, Providence, RI; Southeast Texas Evolutionary Genetics Meeting

Thoracic underreplication predicts minimal *Drosophila* genome size. *Hjelmen CE, Holmes VR, !Burrus C, Johnston JS.
Texas Genetics Society Annual Meeting, College Station, TX (Postdoc Poster Winner); 60th Annual Drosophila Research Conference, Dallas, TX

2018

Markers of Sex Determination in Blow Flies. *†!Jonika MM, Faris AM, Hjelmen CE, Tarone AM.
LAUNCH Undergraduate Research Summer Poster Session, Texas A&M; International Association for Identification, San Antonio, TX; SE Texas Evolutionary Genetics Symposium, Rice University

2017

The impact of regional climate variables on genome size evolution in *Drosophila*. *Hjelmen CE, Johnston JS.
58th Annual *Drosophila* Research Conference, San Diego, CA

2016

How does replication level contribute to genome size evolution in *Drosophila*? *Hjelmen CE, Johnston JS.
The Allied Genetics Conference, Orlando, FL

2015

Transcriptomics must take into account unexpected levels of endoreduplication. *Hjelmen CE, †!Mynes M, Johnston JS.
Arthropod Genomics Symposium, Kansas State University

The rate and pattern of genome size evolution in Drosophilidae and Formicidae. *Hjelmen CE and Johnston JS.
ESA Southwestern Branch Meeting, Tulsa, OK (1st place)

2014

Genome size variation in *Belonocnema treatae*. *Hjelmen CE, Ott JR, Egan SP, Johnston JS.
ESA Meeting, Portland, OR; Arthropod Genomics Symposium, UIUC

The rate and pattern of genome size evolution in Drosophilidae and Formicidae. *Hjelmen CE and Johnston JS.
Arthropod Genomics Symposium, UIUC

Sex, ecology, and the genome in *Belonocnema treatae*. *Hjelmen CE, Ott JR, Egan SP, Johnston JS.
Ecological Integration Symposium, Texas A&M (2nd place)

2013

Extensive variation in genome size among populations in the gall wasp *Belonocnema treatae*. *Hjelmen CE, Ott JR, Egan SP, Johnston JS.
Arthropod Genomics Symposium, University of Notre Dame; ESA Meeting, Austin, TX

TEACHING

Utah Valley University — Department of Biology

Introduction to Bioinformatics (BIOL 1011) — Instructor of Record

Introductory biology topics and their applications in bioinformatics; introductory bioinformatic tools and basic R coding.

- Fall 2021: 2 sections, 22 students
- Spring 2022: 2 sections, 17 students
- Fall 2022: 1 section, 35 students
- Spring 2023: 2 sections, 70 students
- Fall 2023: 2 sections, 49 students
- Spring 2024: 2 sections, 37 students
- Fall 2024: 2 sections, 39 students
- Spring 2025: 2 sections, 34 students
- Fall 2025: 2 sections, 24 students
- Spring 2026: 2 sections, 18 students

Genetics (BIOL 3500) — Instructor of Record

Upper-level genetics: central dogma, regulation of gene expression, genome organization, and population genetics.

- Fall 2021: 31 students
- Spring 2022: 39 students
- Summer 2022: 32 students
- Fall 2022: 41 students
- Spring 2023: 52 students
- Summer 2023: 30 students
- Fall 2023: 47 students

- Summer 2024: 38 students
- Summer 2025: 33 students
- Summer 2026: 39 students

Genomics (BIOL 3550) — Instructor of Record

Sequencing, assembly, annotation, phylogenomics, comparative genomics, and population genomics. Weekly literature discussion and student-driven final projects.

- Fall 2022: 15 students
- Fall 2023: 13 students
- Fall 2024: 13 students
- Fall 2025: 12 students

Phylogenetics (BIOL 4300) — Co-Instructor (with Dr. Heath Ogden)

Tree reconstruction methods: parsimony, maximum likelihood, and Bayesian methodologies. Students generate novel datasets and build reproducible pipelines.

- Fall 2025: 18 students

S-STEM Introduction to Undergraduate Research (BIOL 2900) — Instructor of Record

Professional development, time management, grant management, and research rotations for NSF S-STEM scholarship fellows.

- Fall 2025: 6 students

Principles of Evolution (BIOL 4500) — Instructor of Record

Mechanisms of evolution, sexual selection, Hardy-Weinberg equilibrium, and speciation. Classic literature discussion and student-driven final projects.

- Spring 2023: 10 students
- Fall 2023: 20 students
- Spring 2024: 19 students
- Spring 2025: 27 students
- Spring 2026: 19 students

Molecular Evolution and Bioinformatics (BIOL 4550) — Instructor of Record

Population genetics and genomics, Hardy-Weinberg, neutral theory, and genome sequencing. Semi-weekly primary literature discussions and student-driven final proposals.

- Spring 2024: 15 students

Bioinformatics Capstone (BIOL 4600) — Instructor of Record

Students implement and critique bioinformatic pipelines, completing independent projects with written and oral presentations.

- Spring 2024: 1 student
- Spring 2025: 7 students

Entomology (ZOOL 3430/3435) — Instructor of Record

Insect diversity and classification, anatomy, and physiology. Students curate museum-quality collections identified to family level.

- Fall 2024: 10 students

Evolution of Sex (BIOL 490R-1) — Co-Instructor (with Dr. Jessica Cusick)

Seminar-style course on the evolution of sex from molecular and behavioral perspectives. Students engage with current research and emerging questions.

- Fall 2024: 10 students

Ethical Issues in Biology (BIOL 4260) — Instructor of Record

Discussion-driven course covering current and future ethical issues in biology: AI, genome editing, polygenic risk scores, and more.

- Spring 2026: 13 students

Student Seminar (BIOL 494R) — Instructor of Record

Students focus on writing a term paper on a topic of their choice.

- Fall 2022: 12 students

Texas A&M University — Department of Entomology

The Science of Forensic Entomology (ENTO/FIVS 431) — Lecture TA

Assist in classroom instruction of Forensic Entomology, including collection and identification of insects for courtroom depositions.

- Spring 2017: 100 students

Applied Forensic Entomology (ENTO/FIVS 432) — Laboratory TA

Instruct upper-level students in collection, identification, and curation of entomological materials for forensic investigations.

- Spring 2017: 12 students

Forensic Investigations (FIVS 123) — Online/Distance Learning TA

Online/distance education in forensic evidence evaluation, critical analysis, and scientific methods.

- Spring 2017: 250 students
- Fall 2016: 250 students
- Spring 2016: 430 students

Forensic Implications of Inheritance (FIVS 308) — Laboratory TA

Basic genetics laboratory techniques for upper-level forensics majors: DNA extraction, PCR, and gel electrophoresis.

- Fall 2016: 13 students

Biology of Insects (ENTO 313) — Laboratory TA

Biological aspects of insects, stressing biodiversity; insect curation and identification to order and family level.

- Fall 2015: 30 students

Insect Biodiversity and Biology (ENTO 301) — Laboratory TA

Upper-level majors: curate and identify insects to family using morphological information.

- Spring 2015: 30 students

General Entomology (ENTO 201) — Laboratory TA

Evolutionary and biological importance of insects; identification to order level.

- Fall 2014: 30 students

Guest Lectures

2025

Molecular Techniques in Forensic Entomology
Introduction to Forensic Entomology (ENPP 4573), Oklahoma State University

2021

Genetic Drift | Evolution of Sex Chromosomes | Website Design and Professional Presence Q&A
Evolution (BIOL 4500), Utah Valley University; BIOL 3100, Utah Valley University

Genome Size Evolution [Synchronous Online] | Whole Genome Sequencing Projects [Synchronous Online]
Biotechnology, Rutgers University, Dept. of Entomology

Molecular Techniques in Forensic Entomology [Asynchronous Online]
Forensic Entomology, Texas A&M University, Dept. of Entomology

MENTORED STUDENTS AND DIRECTED LEARNING

Research Students Mentored — (Current Position)

- Joshua Nilsson (Current student)
- Beth Larsen (Current student)
- Haylee Beck (Current student)
- Dawn Hornsby (Current student)
- Bailey Jannuzzi (PhD Student at William & Mary)
- Olivia Frary (PhD Student at University of Rochester)

- Barbara Shelley (PhD Student at University of Utah)
- Sarah Fout (Pharmatech Labs)
- Sam Curnow (Quality Assurance Tester at Caselle)
- Audrey French (Specimen Processor at Myriad Genetics)
- Erick Alvarez (MS Student, University of Utah)
- Remington Motte (PhD Student at University of Arkansas)
- Marissa Ohran (NSF RAMP Post-Bacc Scholar)
- Alex McGuane (Data Analytics and Visualization)
- Michelle Jonika (Postdoctoral Researcher, Heath Blackmon Lab, Texas A&M University)
- Elizabeth Piron (MPH, Epidemiologist)
- V. Renee Holmes (Research Scientist at Lab+ Certified Group)
- Melissa Mynes (Laboratory Technician for Aqua Solutions)
- Margaret Garrett (Medical Student at McGovern Medical School)
- Zoe Ward (Substitute Teacher)
- Anika Sharma (Fulbright PhD Scholar — Received PhD)
- Alli Konstantinov (Current undergraduate)

Student Thesis Committees

2025–2026 Cyrus Nielsen — Honors Project Committee Chair
Title: Contributing to UVU's Insect Teaching Collection and Entomology Student Resources

2025–2026 Austen Miller — Honors Thesis Committee Chair
Title: Simulacra and Speciation

2025–2026 Amaya Roberts — Thesis Committee Member
Title: Field Guide to Grasses of CRNP

2022–2023 Kate Hickman — Honors Thesis Committee Member
Title: Metagenomic Tools for Microbiome Analyses in Non-Model Systems

2022–2023 Kelsey Stone — Senior Thesis Committee Member
Title: Exploring Insect Diversity in Capitol Reef National Park: The Creation of a Field Guide

2022–2023 Avery Larsen — Senior Thesis Committee Member
Title: A Phylogenomic Analysis of Baetidae (Ephemeroptera)

2022–2023 Ernie Vilela — Senior Thesis Committee Member
Title: A Bioinformatic Workflow for Transcriptome Analysis of Opsin Genes in Ephemeroptera

2021–2022 Jeremy Jensen — Senior Thesis Committee Member
Title: The Insects of Capitol Reef: The Creation of a Field Guide

PROFESSIONAL DEVELOPMENT

Certifications and Training

Fall 2022 Online Teaching Academy (25-hour Certification) — Office of Teaching and Learning, UVU
Instructor presence, peer-to-peer interaction, meaningful learning activities, and secure assessments.

Summer 2022 Anti-Racist Pedagogy (25-hour Certification) — Office of Teaching and Learning, UVU
History of racism; supporting anti-racism in the classroom.

Spring 2022 Evidenced-Based Teaching Practices for Higher Education (25-hour Certification) — Office of Teaching and Learning, UVU
Learner-centeredness, backwards design, metacognition, inclusive teaching, assessment, and active learning.

Fall 2021 Teaching First Year Students (12.5-hour Certification) — Office of Teaching and Learning, UVU
Best practices for first-year students: early connections, growth mindset, metacognition.

Learning Circles

Fall 2023 A Voice in the Wilderness (Co-Leader) — Utah Valley University
Spring 2023 Unraveling Faculty Burnout — Utah Valley University
Fall 2022 Everyone is African — Utah Valley University
Spring 2022 On Being a Mentor — Utah Valley University
Fall 2021 Introduction to Critical Race Theory — Utah Valley University

SERVICE

Symposia Organized

2026 Arthropod Genomics and Genome Assembly
International Plant and Animal Genome Meeting, San Diego, CA (Jan. 2026)
Co-organized with Lindsey Perkin (USDA)

2025 The General Entomology Toolbox: Activities and Practices for Improving a Foundational Course
ESA Annual Meeting, Portland, OR (Nov. 2025)
Co-organized with Carly Tribull (Farmingdale State College)

2024 How to Be a Vector for Entomophilia: Infecting the Future
ESA Annual Meeting, Phoenix, AZ (Nov. 2024)
Co-organized with David Serrano (Broward College)

2023 Entomology Education and Experience: How Insects Impact, Inform, and Influence People and Policy
ESA Annual Meeting, National Harbor, MD (Nov. 2023)
Co-organized with David Serrano (Broward College) and Ashleigh Faris (Texas A&M University)

2022 Inspiring the Next Scientists: Entomology as an Educational and Research Tool for Undergraduates
ESA Annual Meeting, Vancouver, Canada
Co-organized with David Serrano (Broward College), Delano Lewis (Burman University), Leah Flaherty (MacEwan University)

Workshops Led

Oct. 2024 / Feb. 2025 Website Workshop for Dept. of Biology, Utah Valley University
General overview of personal/lab websites; GitHub Pages development.

2020 Open Source for Open Science, Texas A&M University [Online]
Basic Statistical Analysis and Visualization in R — 274 registrants

2019 Open Source for Open Science, Texas A&M University
Basic Statistical Analyses in R — 292 registrants

Editorial and Peer Review

Editor Guest Editor, Genes special issue: Genetics of Phenotypic Variation in *Drosophila* and Other Insects (10 articles)

Peer Review (Number of Reviews)

Agricultural and Forest Entomology (2) • Biology (2) • BMC Genomics (2) • Cells (3) • Communications Biology (1) • Diversity (1) • Evolution (1) • Frontiers in Genetics (1) • Genes (5) • Genetics (1) • Genome (3) • Insectes Sociaux (1) • Insects (3) • International Journal of Acarology (1) • International Journal of Environmental Research and Public Health (1) • International Journal of Molecular Sciences (1) • Journal of Heredity (1) • Journal of Insect Science (1) • Journal of Medical Entomology (1) • Mitochondrial DNA Part B (4) • Molecular Biology and Evolution (2) • Molecular Ecology Resources (4) • Plants (1) • Systematic Entomology (2) • Wellcome Open Research (3) •

Western North American Naturalist (1) • PLOS Computational Biology (1) • PLOS Genetics (1) • G3: Genes|Genomes|Genetics (1) • Royal Society Open Science (1) • Scientific Reports (1) • iMeta (1)

General Service

Professional Level

2023–Pres.	ESA FIT (Formal and Informal Teaching) Program Subcommittee
2024	ESA Panelist — Early Career Professional Workshop on Networking
2023	ESA Graduate Student Talk Judge
2023	NSF Reviewer
2019	Southeast Texas Evolutionary Genetics and Genomics—Planning Committee
2016–2017	Chair, SW Branch ESA Student Affairs Committee
2014–2016	Chair, Photo Salon—SW Branch of ESA

University Level

2026	Panelist for Tenure and Midterm Tenure Packets
2024	Speaker, UVU STEAM-CON — Finding your Role Model in STEAM
2024	SCULPT Panelist — Mentoring and Researching with Undergraduates
2023	Co-leader, Faculty and Staff Learning Circles ('A Voice in the Wilderness')
2023	Faculty Panelist, New Faculty Orientation, Utah Valley University
2022	Faculty Panelist, New Faculty Orientation, Utah Valley University
2020	Open Source Open Science Workshop — Instructor and assistant, 274 students
2019	Open Source Open Science Workshop — Instructor and assistant, 292 students

College Level

2024	College Day — Faculty and Student Panel
2022–2026	Darwin Day Committee, Utah Valley University
2021	Dean's Day — Volunteered at Biology Table, Utah Valley University

Department Level

2025–2026	Search Committee Member — Bioinformatics Tenure-Track Position, Dept. of Biology, UVU
2025	Search Committee Chair — Bioinformatics One-Year Lecturer Position, Dept. of Biology, UVU
2024–2025	Search Committee Chair — Animal Biology/Zoology Tenure-Track Position, Dept. of Biology, UVU
2024–2025	Search Committee Chair — Invertebrate Collections Technician Position, Dept. of Biology, UVU
2023–2024	Affinity Groups Mentor — Atheists and Agnostics, Dept. of Biology, UVU
2023	Search Committee — Permanent Lecturer Position, Dept. of Biology, UVU
2021–Pres.	Hiring Committee, Dept. of Biology, UVU (Chair 2022–ongoing)
2021–Pres.	Engagement/Inclusion Committee — Networking Subcommittee; Mentor for Agnostic/Atheist in Science Affinity Group
2021–Pres.	Bioinformatics Degree Program Committee, UVU (Coordinator 2023–ongoing)
2021–2022	Search Committee — Anatomy Tenure-Track Position, Dept. of Biology, UVU
2020	Deposition Panel, Texas A&M University Forensic Entomology Final Project
2019	Deposition Panel, Texas A&M University Forensic Entomology Final Project
2018	AWE (Aggie Women in Entomology) Mentor Panel: The Next Chapter
2018	Postdoctoral Judge, Entomology Mentoring in Research Symposium
2017	AWE (Aggie Women in Entomology) Mentor Panel: The Next Chapter
2017	Postdoctoral Judge, Texas A&M Entomology Graduate Student Forum
2016–2017	President, Entomology Graduate Student Organization — TAMU

2016–2017	Graduate Student Representative, Dept. Faculty Meetings
2015–2017	New Graduate Student Panel
2015–2016	Vice President, Entomology Graduate Student Organization — TAMU
2015–2016	Entomology Representative, Graduate and Professional Student Council
2014–2015	Social Activities Chair, Entomology Graduate Student Organization — TAMU
2014–2017	Texas A&M University Linnaean Games Team

Community Outreach

- Mar. 2025 — Pleasant Grove Library
- Feb. 2025 — Darwin Day, UVU Biology
- June 2024 — Salt Lake Oasis Talk
- June 2023 — Springville Library Outreach
- June 2023 — Spanish Fork Library Outreach
- Feb. 2023 — Darwin Day, UVU Biology
- Mar. 2023 — SHE-roes Event, Thanksgiving Point
- Feb. 2022 — Darwin Day, UVU Biology
- Oct. 2021 — Dean's Day, UVU College of Science
- Feb. 2020 — 'Why are there so many insects in Texas?' for 100 2nd grade students
- Apr. 2017 — SWB ESA Insect Expo
- Mar. 2017 — Family Fun Night, Creek View Elementary
- May 2015 — Creek View Elementary Event; Be the Match Foundation Fundraiser
- Feb. 2015 — SWB ESA Insect Expo
- Oct. 2014 — 'Trunk or Treat' Entomology Halloween Event; 'Boonville Days' Outreach
- May 2014 — Covenant Presbyterian Church Preschool Event

MEMBERSHIPS

- Society for Molecular Biology and Evolution (Active)
- North American Forensic Entomology Association (Active)
- Entomological Society of America (Active)
- Pacific Branch of ESA (Active)
- Beta Beta Beta Biological Honor Society
- Phi Sigma Tau Philosophy Honor Society
- Gamma Sigma Delta Agriculture Honor Society
- Texas Genetics Society (Former) | Society for the Study of Evolution (Former) | Genetics Society of America (Former) | American Society of Naturalists (Former) | Southwest Association of Naturalists (Former) | Southwest Branch of ESA (Former)

AWARDS

2025	UVU Office of Teaching and Learning Travel Award — $600
2025	UVU Grants for Engaged Learning 2-Year Grant — $29,987
2025	Capitol Reef Field Station Colorado Plateau Grant — $7,489
2025	UVU College of Science Dean's Distinguished Faculty Scholarship Award
2024	UVU Scholarly Activities Committee Dissemination Grant — $1,953
2023	UVU Office of Teaching and Learning Travel Award — $700
2022	UVU Office of Teaching and Learning Travel Award — $700
2022	UVU Grants for Engaged Learning (GEL) — $9,766
2019	Texas Genetics Society Annual Meeting Postdoctoral Poster Award Winner
2017	ESA Southwestern Branch 1st Place PhD Oral Presentation
2017	Dept. of Entomology: Award for Outstanding Ph.D. Student

2016	Single-Cell and Gene Expression Grant Award (Fluidigm and TIGGS Shared Molecular-Genomics Workspace)
2016	Dept. of Entomology: Award for Outstanding Ph.D. Student
2015	ESA National Meeting 1st Place Graduate Student Talk — SysEB: Citizen Science, New Methods, and Physiology Section
2015	Vice President for Research: Excellence in Research Award Nomination
2015	Ecological Integration Symposium 1st Place Graduate Student Oral Presentation
2015	Student Research Week 1st Place Graduate Student Oral Presentation
2015	ESA Southwestern Branch 1st Place Ph.D. Poster
2014	Theodore Roosevelt Memorial Fund
2014	Induction into Gamma Sigma Delta Agriculture Honor Society
2014	Ecological Integration Symposium 2nd Place Graduate Student Poster
2013	Texas A&M Merit Fellowship
2013	Induction into Phi Sigma Tau Philosophy Honor Society
2012	Will Rosine Memorial Biology Scholarship
2012	Texas A&M REU EXCITE; Council on Undergraduate Research (CUR) Biology Travel Award
2011	BRIN Research Fellowship; Induction into Beta Beta Beta Biological Honor Society
2010	Sven G. Froiland Biology Scholarship

TECHNICAL SKILLS

Flow Cytometry • Genome Size Estimation • Phylogenetics • Western Blots • DNA and RNA Extraction • Model Insect Husbandry • Insect Curation and Identification • R (Statistics and Data Visualization) • Genomic Analyses (Local and HPC) • Genome Assembly • Transcriptomic Analysis • Sequence Alignments • Comparative Phylogenetics

REFERENCES

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